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Gurgaon Apartment Market Analysis using Real Estate Listings

Project Domain

Power BI

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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3. Problem Statement

The real estate sector generates large volumes of property listing data containing information such as property prices, locations, configurations, amenities, and property characteristics. However, this data is often collected from multiple online platforms and exists in raw, inconsistent, and poorly structured formats. As a result, buyers, investors, and analysts face difficulty comparing property prices, identifying trends, and making data-driven decisions.

In rapidly growing cities like Gurgaon, the real estate market is highly dynamic, with significant variations in property prices across sectors, societies, and apartment configurations. Without proper data cleaning, analysis, and visualization, valuable insights from property listings remain hidden and underutilized.

This project aims to address this challenge by transforming raw Gurgaon apartment listing data into a structured dataset and developing an interactive real estate analytics dashboard using Excel and Power BI. The project will demonstrate how data cleaning, business intelligence tools, and AI assistance can help uncover pricing patterns, investment opportunities, and market trends.

The insights generated will help potential buyers, investors, and analysts better understand the Gurgaon real estate market and make informed decisions based on data-driven analysis.

4. Objectives

The key objectives of this project are:

• To collect and explore a dataset containing residential apartment listings in Gurgaon.

• To understand the structure, attributes, and quality of the dataset including fields such as price, property size, location, configuration, and amenities.

• To clean and preprocess the raw data using Power Query by removing duplicates, handling missing values, correcting encoding issues, and standardizing formats.

• To perform exploratory data analysis using Excel and Power BI to identify key real estate market trends.

• To design and build an interactive Real Estate Sales and Property Portfolio Dashboard using Power BI.

• To generate key business KPIs such as average price, price per square foot, and sector-wise property distribution.

• To use AI tools for documentation support, data explanation, and insight generation.

• To demonstrate how business intelligence tools can transform raw property listing data into actionable market insights.

5. Methodology / Implementation Plan

The project will follow a structured five-phase workflow:

#	Phase	Description
1	**Data Collection & Exploration**	The project begins with obtaining a dataset containing Gurgaon apartment listings. The dataset includes fields such as property name, price, area, number of bedrooms, bathrooms, balconies, society name, and address. The dataset will be explored to understand its structure, number of records, and data types.
2	**Data Cleaning & Preprocessing**	The dataset will be cleaned using Excel Power Query. This includes removing duplicate records, correcting encoding errors, extracting numeric values from text fields, and standardizing columns such as price, area, and property configuration. Unnecessary columns will be removed to prepare the dataset for analysis.
3	**Analysis & AI Integration**	After cleaning the dataset, exploratory analysis will be performed using Excel formulas and Power BI measures. Key metrics such as average property price, price per square foot, and property distribution by configuration will be calculated. AI tools will be used to assist in generating summaries and interpreting the results.
4	**Visualization / Output**	An interactive Power BI dashboard will be created to visualize key real estate insights. Charts, KPI cards, and filters will allow users to explore the dataset dynamically. Visualizations will include sector-wise price comparison, BHK distribution, and price-per-square-foot analysis.
5	**Documentation & Submission**	The final stage includes documenting the entire project process, preparing the project report, uploading files to GitHub, and recording a demonstration video showing the working dashboard and insights.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Microsoft Copilot — via Microsoft 365 subscription — where applicable

- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

The dataset used for this project consists of residential apartment listings in Gurgaon, Haryana. The dataset contains over 500 property records and includes attributes such as property name, society name, price, area, bedroom count, bathroom count, balcony count, facing direction, and property age.

The dataset was obtained from publicly available property listing sources and converted into CSV format for analysis. Data cleaning was performed using Excel Power Query to ensure the dataset is accurate and suitable for analytical processing.

C. Project Timeline

Phase	Task	Completion
Phase 1	Dataset collection and understanding	Week 1
Phase 2	Data cleaning and preprocessing	Week 2
Phase 3	Data analysis and KPI generation	Week 2–3
Phase 4	Dashboard development in Power BI	Week 3
Phase 5	Report writing and project submission	Week 4

7. Expected Outcomes / Conclusion

The project is expected to produce a structured and cleaned real estate dataset that can be used for meaningful analysis and visualization. By applying business intelligence techniques using Excel and Power BI, the project will generate insights into Gurgaon's residential apartment market.

The final output will include an interactive dashboard presenting key metrics such as average property price, price per square foot, sector-wise price comparison, and property

configuration trends. These insights can help stakeholders better understand real estate market dynamics.

Additionally, the project will demonstrate how AI tools can assist in data interpretation, documentation, and insight generation, highlighting the role of AI in modern data analytics workflows.

8. References

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